

Fuzzing with Code Fragments

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Fuzzing with Code Fragments

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Fuzz testing is an automated technique providing random data as input to a software system in the hope to expose a vulnerability. In order to be effective, the fuzzed input must be *common enough* to pass elementary consistency checks; a JavaScript interpreter, for instance, would only accept a semantically valid program. On the other hand, the fuzzed input must be *uncommon enough* to trigger exceptional behavior, such as a crash of the interpreter. The *LangFuzz* approach resolves this conflict by using a *grammar* to randomly generate valid programs; the code fragments, however, partially stem from *programs known to have caused invalid behavior before*. LangFuzz is an effective tool for security testing: Applied on the Mozilla JavaScript interpreter, it discovered a total of 105 new severe vulnerabilities within three months of operation (and thus became one of the top security bug bounty collectors within this period); applied on the PHP interpreter, it discovered 18 new defects causing crashes.

Christian Holler, Mozilla Corporation

Kim Herzig, Saarland University

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